

The leading Mellin (zeta) coefficient of a normal-crossing amplitude

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Abstract

Unit 222 is Astra #24's optional programme E2. The paper evaluates normal moments through the zeta function $\zeta(z) = \int (u^{2k})^z \eta(u) u^h du$ (transform of $K = u^{2k}$), whose leading Laurent coefficient at the pole $z = -\lambda$ of order m is $a_{-m} = \prod_{i \in J} \frac{1}{2k_i} \prod_{i \notin J} \frac{1}{h_i + 1 - 2k_i \lambda}$ at $b = 1$ (eq. `a_minus_m_explicit`). We prove the Abelian, real-axis form for a continuous amplitude on the unit box:

$$s^m \int_{(0,1]^d} \eta(u) \prod_i u_i^{2k_i(-\lambda+s)+h_i} du \longrightarrow \prod_{i \in J} \frac{1}{2k_i} \int \eta(\pi u) \prod_{i \notin J} u_i^{h_i - 2k_i \lambda} du \quad (s \rightarrow 0^+)$$

(`mellin_leading_tendsto`), with π zeroing the minimising coordinates. For $\eta = 1$ the limit is a_{-m} (`mellinCoeff_one`); for equal ratios it is $\eta(0) \prod_i \frac{1}{2k_i}$ (`mellinCoeff_equal`); and the Laplace-side coefficient of Headline VIII equals $\Gamma(\lambda) \beta^{-\lambda} / (m-1)!$ times the Mellin coefficient (`amplitudeCoeff_eq_gamma_mul_mellinCoeff`) — the paper's “single $\Gamma(\lambda)$ factor from the Laplace–Tauberian passage”, obtained here by comparing two Abelian limits. Not a meromorphic continuation or complex pole theorem. Zero `sorry/axiom`; 9099 jobs.

1 The things formalised

```
def mellinConst h k l := i, if ratioExp h k i = 1 then 1/(2 k) else 1
def mellinCoeff h k l :=
  mellinConst h k l * u in unitBox d, (faceProj u) * residualWeight u
def mellinWeight h k l N u := if 0 < N then ( i, u ^ (2k (-l + 1/N) + h)) / N ^ m else 0
theorem integral_monomial_mellinWeight :
  ( u ^ ) * mellinWeight N = i, mellinFactor N i
theorem mellinFactor_tendsto : mellinFactor N i → mellinFactorLim i (N → ∞)
theorem mellin_face_moment_eq / _eq_zero : face moments of mellinConst * residualWeight
theorem mellinWeight_tendsto : u * mellinWeight N u → mellinCoeff h k l
theorem mellin_leading_tendsto : Tendsto (fun s => s ^ m * u * u ^ (2k (-l+s)+h))
  ([>] 0) ( (mellinCoeff h k l) )
theorem mellinCoeff_one : mellinCoeff h k l 1 = i, if J then 1/(2k) else 1/(h+1-2k l)
theorem mellinCoeff_equal (hratio) : mellinCoeff h k l = 0 * i, 1/(2 k)
theorem amplitudeCoeff_eq_gamma_mul_mellinCoeff :
  amplitudeCoeff h k l = Γ l * ^(-l) / (m-1)! * mellinCoeff h k l
```

2 Mathematics

Same skeleton as the Laplace-side amplitude theorem (unit 187): the monomial-moment transfer `tendsto_integral_of_monomials` reduces a continuous amplitude to monomials, and for the Mellin weight the monomial moments are *closed forms*: $\int_0^1 u^a du = 1/(a+1)$, so

$s^m \int u^\gamma \prod u_i^{2k_i(-\lambda+s)+h_i} = \prod_i \frac{1}{\gamma_i+e_i(s)+1} / \prod_{i \in J} (1/s)$ with $e_i(s) = -1 + 2k_i s$ on J . A minimising coordinate with $\gamma_i = 0$ contributes $\frac{1}{2k_i s} \cdot s = \frac{1}{2k_i}$; with $\gamma_i > 0$ it contributes $\frac{s}{\gamma_i+2k_i s} \rightarrow 0$ (the face moment vanishes); a nonminimal coordinate contributes $\frac{1}{\gamma_i+h_i+1-2k_i \lambda+2k_i s} \rightarrow \frac{1}{\gamma_i+h_i+1-2k_i \lambda}$. No Gamma functions, no logarithms: the $\Gamma(\lambda)/(m-1)!$ of the Laplace side is visible as the ratio of the two limiting functionals. Numerics: $d = 2$, $h = (0, 1)$, $k = (1, 1)$, $\eta = 1 + u_1$: $s V(-\frac{1}{2} + s) = 0.6439, 0.7377, 0.7488$ at $s = 10^{-1}, 10^{-2}, 10^{-3}$ against $3/4$; equal ratios $h = 0$, $\eta = 1 + 0.3u_0 + 0.2u_1^2$: $s^2 V = 0.2670, 0.2520, 0.2502$ against $1/4$.

3 Lean notes

- Guard the weight (if $0 < N$ then ... else 0) so that the transfer lemma's `N, IntegrableOn` hypothesis holds without a positivity side condition.
- `Finset.prod_ite` would not elaborate inside `rw`; $\leftarrow \text{Finset.prod_filter } p \ f$ with explicit arguments gives $(\text{if } p \ i \text{ then } N \text{ else } 1) = i \text{ filter } p, N,$ then `Finset.prod_const` and `multCount_eq_card`.
- `Finset.prod_div_distrib` inside `rw` hits the first product (the left-hand $\prod 1/a_i$); `simp only` normalises both sides instead.
- Box membership is `i ∈ Set.univ (Set.mem_univ i, not Finset.mem_univ)`.
- $s \rightarrow 0^+$ from $N \rightarrow \infty$: compose with `tendsto_inv_nhdsGT_zero` and rewrite $1/s^{-1}, (s^{-1})^m$ pointwise.

4 Status

E2 done (Headline XXI). All Astra #24 items now complete (H2-lite XX-XX'' + prior lemmas, E2). Freeze; a review v17 of units 221–222 is the natural closing step.